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The Effects of Service Recovery Actions on Customers' Post-Recovery Responses to Online Travel Agencies (OTAs): The Moderating Role of Price

Jungjin Hwang 

Tourism Studies, Kyungnam University, Changwon-si, Republic of Korea

Correspondence: Jungjin Hwang (jamescrow@kyungnam.ac.kr; emptiness96@gmail.com)**Received:** 2 November 2023 | **Revised:** 29 April 2024 | **Accepted:** 5 August 2024**Keywords:** justice theory | online travel agency | service failure | service recovery

ABSTRACT

The main purpose of this study is to investigate the impacts of service recovery actions on customers' post-recovery responses to initial service failure and a double deviation and to investigate the moderating effect of price on the relationship between recovery actions and these responses. The findings show that active recovery actions led to more favorable post-recovery responses than passive actions in both initial and double deviation situations. No moderating effect of price was found between recovery actions and customers' post-recovery responses. From an academic perspective, this study helps researchers understand how customers positively perceive recovery actions that reflect procedural and interactional justice in service failure situations. The findings also suggest that practitioners of OTAs should engage in active recovery actions that include prompt communication on the progress of the incident (i.e., procedural justice) and a sincere apology (interactional justice) to alleviate customers' negative responses due to service failures.

1 | Introduction

Advances in internet technology have changed the way tourists book tour products and bolstered the proliferation of online booking channels, such as online travel agencies (OTAs) (Rianthong, Dumrongsiri, and Kohda 2016). The importance of online travel websites continues to grow as tourists increasingly try to arrange the exact details they desire on their trips through the best means possible (Chen and Tussyadiah 2021). Through reviews provided by OTAs, travelers can obtain information about travel experiences they may experience or ask questions and find answers about the destination they plan to visit (Oriade and Robinson 2019). For tourists, booking tourism services through OTAs provides certain advantages, such as convenience of use, cost reduction, time saving, and product diversity (Hao et al. 2015; Liu and Zhang 2014; Pappas 2017). However, online transactions made through OTAs are more difficult to control than those made by offline businesses because, in the

case of the former, product buyers and sellers are physically separated (Ahmad 2002; Zhao et al. 2014). Thus, service failures occur more frequently in an online service environment than in face-to-face circumstances (Harris et al. 2006). Furthermore, online companies generally experience difficulties in building mutual trust with customers because there is a measure of uncertainty that comes with such situations, as well as a lack of control (McKnight and Chervany 2001). If a service failure does occur, and the provider cannot recover in time, customers' trust can quickly erode (Zhao et al. 2014). Thus, because service failures on the part of online companies can cause negative reactions from customers, effective service recovery actions remain critical to the success of such enterprises (Lin et al. 2018; Sreejesh, Anusree, and Abhilash 2019).

While there has been considerable research on service failures and the effectiveness of recovery actions on customers' behavioral outcomes in the context of traditional offline service

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companies, there is a dearth of research on this topic in the context of OTAs (Manu, Sreejesh, and Paul 2021; Wei 2021; Zhao et al. 2014). Moreover, previous research has studied the effectiveness of service recovery actions using procedural and interactional justice, which are those dimensions of justice theory known to be most effective in addressing service recovery in offline service environments (Bacile et al. 2018; Chapman and Chapman 2016; Crisafulli and Singh 2017; Del Río-Lanza, Vázquez-Casielles, and Díaz-Martín 2009; Liu et al. 2019; Wang et al. 2011). However, few studies of service failures in OTA contexts have been conducted, and those that have generally do not attempt to measure the effectiveness of service recovery actions in terms of those two factors. In addition, existing studies on service failures and recovery actions in the context of online service companies have focused mainly on how recovery actions for service failures affect customers' post-recovery responses (Manu, Sreejesh, and Paul 2021; Wei 2021; Zhao et al. 2014). Accordingly, these studies have neglected to investigate how affected customers react when problems related to service failures are either resolved or unsettled after service recovery actions (i.e., double deviation).

Double deviation in this context refers to additional failures in recovery actions that occur after an initial service failure and implies that the customer's expectations of a smooth service experience were deviated from on at least two occasions (Bitner, Booms, and Tetreault 1990). Because double deviation can lead to extremely dissatisfied customers displaying retaliatory behavior (Edvardsson, Tronvoll, and Höykinpuro 2011; Khamitov, Grégoire, and Suri 2020; Lee et al. 2021), service providers need to prepare effective recovery actions in advance of a potential second service failure. Generally, studies examining customers' reactions to recovery actions during a double deviation in OTAs are exceedingly rare. To address this research gap, this study first examines how customers' post-recovery responses to service failures involving OTAs vary depending on the extent to which the recovery actions are active or passive, which relates to the notions of procedural and interactional justice. Furthermore, this study investigates how different degrees of recovery affect customers' post-recovery responses in double-deviation situations.

In relation to utility theory, Lancaster (1971) argued that customers purchase products or services with the expectation of a certain level of product or service quality and at a certain price range to obtain utility. Customers' expectations of service quality are generated based on tangible or intangible factors and implicit service promises, such as price (Zeithaml, Berry, and Parasuraman 1993). Overall, customers employ price as a main indicator of quality when a product or service is expensive (Crisafulli and Singh 2017). Thus, if customers pay a higher price for a hotel room, they expect a higher level of service commensurate with the price paid (Anderson, Fornell, and Lehmann 1994). Customers who experience dissatisfaction with a service tend to complain more when the price of the service is high (Best and Andreasen 1976).

From these prior studies, it is easy to infer that because customers have higher expectations of service when the price is higher, the expectations of recovery actions for service failures are also higher when the price is high. Depending on the different levels

of expectations for recovery actions, customers' post-recovery responses will also manifest differently. In other words, price is expected to moderate the impacts of recovery actions for service failures on customers' post-recovery responses. Despite the important influence that price has on recovery actions and post-recovery responses, previous literature on service failures and recovery actions has neglected to use price as a key variable (Kim, Lee, and Mattila 2014). Therefore, this study also examines how high and low prices affect the relationship between service recovery actions and customers' post-recovery responses.

In summary, this study examines (a) the impact of recovery actions in connection with procedural and interactional justice on customers' post-recovery responses; (b) the influences of recovery actions related to these two types of justice on the customers' post-recovery responses in the context of double deviation; and (c) the moderating effects of price on the relationship between recovery actions and post-recovery responses for OTAs. The findings from this study have important implications for academic research and industry related to OTAs. From an academic perspective, the results of this study serve to address various research gaps related to the summary of the literature presented above. In addition, the results of this study may provide scholars planning new research endeavors with new and perhaps more fruitful directions for their work. From a practical perspective, practitioners of OTAs struggle to deal with the negative emotions and attitudes of customers who have experienced service failure due to a lack of effective strategies, which can be attributed to a lack of research on the effectiveness of service recovery actions. Thus, it is hoped that this study can improve our understanding of the impacts of recovery actions regarding procedural and interactional justice on customers' post-recovery responses in the context of initial service failure and double deviation. In doing so, the findings may help OTA practitioners develop more effective recovery strategies to recover from such failures and create opportunities to mitigate customers' negative post-recovery responses in these situations.

2 | Literature Review

2.1 | Service Failures

Service failures are inevitable because of the many uncontrollable elements in the service industry and in customers' unspecified perceptions of services (Berkes and Davidson-Hunt 2007; Zhao et al. 2014). Service failure is defined as the gap between a customer's expectation of a service and his or her perception of the service outcome (Gronroos 1988; Holloway and Beatty 2003). In other words, service failure is the service provider's incapacity to offer a basic or core service, resulting in a loss of utility for the customer (Smith, Bolton, and Wagner 1999).

Attribution theory provides a theoretical background for understanding customers' perceptions of service failures. Customers tend to assess the causes of a service failure along three dimensions: the locus of causality, controllability, and stability. The locus of causality refers to the customer's attempt to determine who is responsible for a service failure after it is experienced (Manu, Sreejesh, and Paul 2021). In the case of service failures involving a company, customers may believe they are entitled

to an apology or refund (Weber and Hsu 2022). Controllability reflects the customer's perception of whether the service provider could have prevented the service failure (Lu, Mody, and Andajigarmaroudi 2020). Stability evaluates the degree to which the cause of the service failure is permanent or transient (Folkes 1988). When the cause of a service failure is stable, the same failure is more likely to occur, which differs from circumstances in which the cause is unstable, the result of which could be more serious problems for the firm (Hwang, Gao, and Mattila 2020).

2.2 | Service Recovery

Previous studies have suggested that customers who experience service failure expect fair treatment from service providers (Nikbin et al. 2010; Orsingher, Valentini, and De Angelis 2010). Such perceptions related to fairness influence their satisfaction with respect to service recovery efforts (Mattila 2001; McCollough, Berry, and Yadav 2000; Smith, Bolton, and Wagner 1999). The fairness of recovery actions is closely related to customers' perceptions of justice in the evaluation of service failure (Gustafsson 2009). Accordingly, justice theory provides a comprehensive theoretical framework for understanding how customers perceive service failure, recovery actions, and the complaint process (Tax, Brown, and Chandrashekar 1998).

According to justice theory, customers usually evaluate the fairness of service recovery actions from three perspectives: distributive, procedural, and interactional justice (McColl-Kennedy and Sparks 2003). Distributive justice refers to customers' perceptions of fairness regarding compensation, which consists of the actual outcome provided to customers in the process of the recovery action (Xu, Liu, and Gursoy 2019). Procedural justice is closely related to customers' perceptions of fairness regarding the policies and procedures used in the decision-making surrounding complaint management (Lind and Tyler 1988) and is evaluated by a firm's speed of response to service failures identified by customers (Liao 2007). Interactional justice denotes customers' perceptions of how fair an interpersonal interaction in the recovery process may have been (Blodgett, Hill, and Tax 1997; Swanson and Hsu 2011); an apology is most often associated with customer-perceived interactional justice (Goodwin and Ross 1992; Smith, Bolton, and Wagner 1999).

The previous literature indicates that procedural and interactional justice are more effective than distributive justice in eliciting customers' positive recovery responses to service failure under certain situations (Bacile et al. 2018; Chapman and Chapman 2016; Crisafulli and Singh 2017; Del Río-Lanza, Vázquez-Casielles, and Díaz-Martín 2009; Liu et al. 2019; Wang et al. 2011). Scholars have pointed out that when a customer experiences a critical service failure, the perception of justice is influenced by the speed of the response from the service provider (i.e., procedural justice) and by how they are treated in this process (i.e., interactional justice) (Bacile et al. 2018; Crisafulli and Singh 2017). Liu et al. (2019) showed that when customers perceive that a service failure is severe, a prompt response (i.e., procedural justice) leads to higher post-recovery satisfaction than monetary compensation alone (i.e., distributive justice). In a similar vein, Wang et al. (2011) argued that interactional justice

provides customers with the most successful service recovery outcomes. Because this study seeks to investigate the effects of service recovery actions in severe service failure situations, the focus here is on examining the impacts of recovery actions involving both procedural and interactional justice.

2.3 | Active and Passive Service Recovery Actions

Service recovery has been defined as a series of actions performed by a firm to resolve customer complaints caused by a service failure (Gronroos 1988). The most widely used model for comprehending customers' responses to service recovery in the hospitality industry is the disconfirmation paradigm (Bearden and Teel 1983; Cai and Chi 2021; Oliver 1980). According to this model, customers seek to compare their perceived service performance with their expectations of the service (Oliver 1980). In other words, customers compare the service recovery actions actually taken by service providers to the expectations they have that the service providers will be able to correct the service failures (McCollough, Berry, and Yadav 2000). Thus, the disconfirmation paradigm suggests that to increase customer satisfaction with recovery performance in the event of a service failure, adequate recovery tactics that can actually meet customer expectations are essential. In this sense, prior studies have found that effective recovery tactics positively affect post-recovery outcomes (e.g., satisfaction, word of mouth, repatronage intention, etc.) (Kim, So, and Mihalik 2022; Weber and Hsu 2022).

As mentioned above, when the degree of service failure is severe, the extant research shows that recovery actions relating to procedural and interactional justice have more positive impacts on customers' post-recovery responses than those related to distributive justice (Bacile et al. 2018; Chapman and Chapman 2016; Crisafulli and Singh 2017; Del Río-Lanza, Vázquez-Casielles, and Díaz-Martín 2009; Liu et al. 2019; Wang et al. 2011). In addition, scholars claim that the simultaneous interaction of both procedural and interactional justice is necessary to increase customer satisfaction with recovery actions in the event of a service failure, and if either is missing, the customer will not be satisfied with those actions (Chen, Hyun, and Lee 2022; Greenberg 1990a,b; McCollough, Berry, and Yadav 2000; McFarlin and Sweeney 1992; Tax, Brown, and Chandrashekar 1998). Among the two types of justice considered here, scholars argue that recovery actions relevant to procedural justice (e.g., prompt responses) lead to the most positive customer post-recovery responses (Chapman and Chapman 2016; Del Río-Lanza, Vázquez-Casielles, and Díaz-Martín 2009; Liu et al. 2019).

This study employs a scenario-based experimental design based on the assumption that procedural and interactional justice serve as stimuli for recovery actions to examine the effects of these approaches on customers' post-recovery responses. Specifically, the present study investigates the impacts of recovery actions on recovery outcomes by manipulating such actions into active and passive categories and by focusing more on the degree to which the recovery actions are related to procedural justice. Those recovery actions performed in an active way indicate that the service staff have actively identified the cause of the customer's complaint, acted proactively to resolve the issue(s), and genuinely apologized for any service failures. In contrast,

passive recovery actions would imply that service staff are slow to listen to customer complaints, slow to resolve issues that cause complaints, and that their apologies for service failures are insincere.

2.4 | Effects of Active and Passive Service Recovery Actions

According to Fishbein and Ajzen (1975), an attitude is defined as “a learned predisposition to respond in a consistently favorable or unfavorable manner with respect to a given object” (p. 10). Other scholars have referred to the notion as “a psychological tendency that is expressed by evaluating a particular entity with some degree of favour or disfavour” (Eagly and Chaiken 1993, 1). In this study, to accurately reflect the research context, an attitude is defined as a customer’s psychological tendency that is expressed by assessing recovery actions related to the service failure of OTAs in a consistently favorable or unfavorable manner.

Along with attitude, another important variable that predicts customers’ responses toward a company is satisfaction (Oliver and Bearden 1985). In the hospitality and tourism industry, satisfaction is considered an important factor for service providers because it leads to increased customer loyalty, ongoing customer relationships, and improved growth and profitability (Goeldner and Ritchie 2007). Satisfaction in this context has been defined as a customer’s reaction to a perceived discrepancy between prior expectations of a product or service and its actual perceived performance after using it (Zeithaml, Berry, and Parasuraman 1993). Simply put, satisfaction refers to a customer’s emotional and psychological responses based on a subjective assessment of service performance (Oliver 1980). Therefore, in the present study, satisfaction refers to a customer’s emotional and psychological reactions to perceived discrepancies between prior expectations and actual perceived performance after experiencing recovery actions.

The last dependent variable was purchasing intention. According to the theory of reasoned action, behavioral intention is a criterion for predicting actual behavior (Fishbein and Ajzen 1975). Ajzen (1991) defined such intentions as “indications of how hard people are willing to try, of how much of an effort they are planning to exert, in order to perform the behaviour” (p. 181). Following this perspective, purchase intentions are defined here as an indicator of how much effort a customer plans to exert to purchase a service from the company that provides the service recovery actions.

According to attribution theory, when a service failure occurs, customers generally try to determine who is responsible for the failure (Manu, Sreejesh, and Paul 2021). If the cause of the failure lies in the company, customers expect appropriate recovery actions from the company (Weber and Hsu 2022). Existing scholars have found that recovery actions relevant to procedural and interactional justice, among all three dimensions of justice theory, elicit more positive post-recovery responses from customers than those regarding distributive justice (Bacile et al. 2018; Chapman and Chapman 2016; Crisafulli and Singh 2017; Del Río-Lanza, Vázquez-Casielles,

and Díaz-Martín 2009; Liu et al. 2019; Wang et al. 2011). From the perspective of the disconfirmation paradigm, the results of previous research show that recovery actions related to procedural and interactional justice incited positive post-recovery responses from customers, thereby indicating that these recovery actions met the customers’ expectations for service recovery (McCollough, Berry, and Yadav 2000; Oliver 1980). Generally, the existing literatures have found that effective recovery actions have a positive effect on customers’ post-recovery responses (e.g., attitudes, satisfaction, word of mouth, re-patronage intention, etc.; Bacile et al. 2018; Chapman and Chapman 2016; Crisafulli and Singh 2017; Del Río-Lanza, Vázquez-Casielles, and Díaz-Martín 2009; Kim, So, and Mihalik 2022; Liu et al. 2019; Wang et al. 2011; Weber and Hsu 2022). Prior scholars have also claimed that more active service recovery actions reduce customers’ negative emotions toward the company and enhance positive emotions (Clemmer and Schneider 1996; Xu, Liu, and Gursay 2019). Qiu et al. (2021) confirmed that as customers’ trust in OTAs increases, they show a higher level of engagement to the company’s brand or the company itself.

Based on the findings from previous studies regarding service recovery actions, this study postulates that active recovery actions related to procedural and interactional justice will lead to more favorable post-recovery responses (i.e., attitudes toward a company, satisfaction, and purchase intentions) than passive recovery actions for these two types of justice (Folger and Konovsky 1989; Greenberg 1990a,b; Grégoire, Tripp, and Legoux 2009; McFarlin and Sweeney 1992; Wirtz and Mattila 2004; Xu, Liu, and Gursay 2019). The following hypotheses were thus developed:

H1. *Customers will show more favorable attitudes toward a company when it performs service recovery actions in an active way (compared to a passive way) in the case of service failure.*

H2. *Customers will be more satisfied with a company when it performs service recovery actions in an active way (compared to a passive way) in the case of service failure.*

H3. *Customers will show a more favorable intent to purchase from a company when it performs service recovery actions in an active way (compared to a passive way) in the case of service failure.*

2.5 | Effects of Active and Passive Recovery Actions on Double Deviation

While this study examines customers’ post-recovery responses to active and passive recovery actions for service failures, it also investigates how the recovery actions affect customers’ post-recovery responses according to the different consequences of those actions. Here, the different consequences of the service recovery actions refer to whether the service failure has been fully recovered or not been resolved. As mentioned earlier, existing studies claim that the more positively customers evaluate service recovery actions in terms of their procedural and interactional justice (i.e., active recovery actions), the more favorable customers’ post-recovery responses

will be (i.e., compared to passive recovery actions; Folger and Konovsky 1989; Greenberg 1990a,b; McColl-Kenney and Sparks 2003; McFarlin and Sweeney 1992; Xu, Liu, and Gursoy 2019). In addition, prior literature suggests that service employees' positive attitudes and behaviors (i.e., active recovery actions) can give customers the perception that employees will do anything within their power to recover from service failures, which in turn alleviates customers' negative emotions (Clemmer and Schneider 1996).

In relation to these claims, Xu, Liu, and Gursoy (2019) confirmed that the positive attitudes and behaviors of service staff reduce customers' negative emotions regarding service failures and increase their positive emotions. Based on these arguments, it is easy to assume that when a service failure is fully recovered, customers will show more favorable post-recovery responses to the service provider when they experience active recovery actions compared to passive actions. Even when a service failure is not fully recovered (i.e., double deviation), it can still be assumed, although to a lesser extent, that customers who experience active recovery actions will show more favorable post-recovery responses than those who perform passive actions. Thus, the following hypotheses were developed:

H4. *Customers will show more favorable post-recovery responses toward a company when it recovers from a service failure through activity compared to passivity.*

H5. *Even if a service failure is not fully recovered, customers will show more favorable post-recovery responses toward a company when it undertakes service recovery actions actively compared to passively.*

2.6 | Effects of Price on Active and Passive Recovery Actions

Price is another important factor in determining customers' expectations of service quality (Anderson, Fornell, and Lehmann 1994). According to social exchange theory, customers judge service failure experiences in relation to a service

purchased at a high price compared to a service purchased at a low price as a more unfair exchange (Emerson 1976; Lee et al. 2023). Thus, customers are likely to have more negative emotions in response to service failures regarding the service products they purchased at higher prices than those purchased at lower prices (Smith, Bolton, and Wagner 1999). Xu, Liu, and Gursoy (2019) found, for example, that customers who purchased plane tickets at a higher price had more negative emotions in response to service failures than those who purchased tickets at a lower price.

To consider another related example, because of the higher expectations of customers who book hotel rooms at relatively high prices, they also have higher expectations of the hotel's recovery actions in response to service failures than their counterparts. Thus, customers who purchase hotel rooms at relatively high prices will have significantly more unfavorable post-recovery responses to passive recovery actions than those who purchase them at low prices. In the case of active recovery actions, the difference in favorability related to post-recovery responses between customers who purchased hotel rooms at relatively high prices and those who purchased them at low prices would not be as great as in the case of passive recovery actions. This is because customers who purchase hotel rooms at low prices will show higher favorability for active recovery actions than their counterparts due to their lower expectations of recovery actions. Conversely, customers who purchase hotel rooms at relatively high prices will show lower favorability for active recovery actions than their counterparts because they have higher expectations for recovery actions in general. As a result, this study postulates that room rates will moderate the impact of service recovery actions (whether active or passive) on customers' post-recovery responses. Thus, the following hypothesis was proposed:

H6. *Customers who purchase hotel rooms at relatively low prices will exhibit more favorable post-recovery responses to active and passive recovery actions for service failures than those who purchase hotel rooms at relatively high prices.*

The following Figure 1 depicts the conceptual framework of this study (Figure 1).

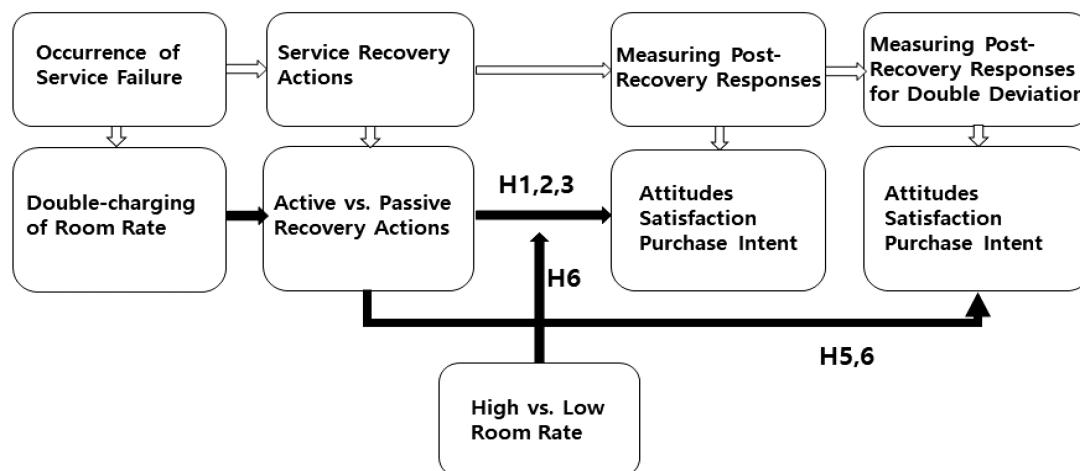


FIGURE 1 | Conceptual framework.

3 | Methods

3.1 | Research Design

This study employed a randomized 2 (service recovery actions: active vs. passive) $\times$ 2 (price: high vs. low) full factorial design to test the hypotheses. A scenario-based experiment was selected because experimental scenarios allow respondents to play themselves in familiar roles and have been proven to produce valuable insights into individual attitudes and intentions regarding the service failure and recovery process (Cranage and Mattila 2006; Greenberg and Eskew 1993). In this case, respondents were asked to read a scenario about the service failure of an OTA in processing a hotel room rate charge. Previous studies claim that one of the major failures that can occur in the provision of an online service is a payment barrier, such as double charging (Holloway and Beatty 2003).

This study employed a fictitious online travel agency named “OTA.com.” In the scenario, the participants were asked to imagine that they had gone on a road trip and had booked a hotel room through OTA.com. Hotel room rates were manipulated by depicting a higher price or lower price compared to similar hotels in the area. The room rates were paid to OTA.com at the time of the reservation. About a week after the road trip, they confirmed that the hotel had also charged the same room rates. They called OTA.com to complain about the problem.

Service recovery actions performed in an active way were described here as the service representative's active efforts to recover from the service failure situation. For example, the staff in this case were described as trying to resolve the problem by contacting the hotel and calling the customer to confirm that the hotel had taken proper actions regarding the problem within the time agreed upon with the customer. The staff then genuinely apologized for the service failure.

In contrast, service recovery actions performed in a passive way were depicted as the service representative taking passive actions in relation to the service failure. For instance, the staff in this case would not have attempted to ascertain the details of the customer's complaint and would not agree to contact the customer again after receiving confirmation from the hotel about the problem. The staff's apologies for the service failure would thus be seen as insincere. All other elements besides the level of service recovery actions and price were the same across all iterations of this scenario.

To test H4 and H5 regarding the differences in the impacts of active and passive service recovery actions on customers, specifically with respect to their post-recovery responses to circumstances wherein the service failure had or had not been recovered (i.e., double deviation), two different short scenarios were added to the questionnaires. The scenario describing the recovery of the service failure included the double-charged room rate eventually being refunded, and the staff calling the customer to confirm the refund. Conversely, the scenario that describes a situation in which the service failure was not fully resolved included the service staff repeatedly telling the customer that the refund process was delayed and they needed to

wait a bit longer. In the real world, when service recovery efforts fail, it is common practice for customers to be kept informed of delays in the refund process, which is why the above scenario was created.

3.2 | Data Collection

Participants were recruited from a large university in the north-eastern United States. College students were chosen because they have the highest frequency of internet use compared to other generations and are considered an important cohort for online retailers (Jai, Burns, and King 2013; Larose and Rifon 2007; National Retail Federation 2007). In existing studies related to online shopping behaviors, college students have often been used as research samples (Agag and El-Masry 2016; Ashraf, Thongpapanl, and Auh 2014; Jai, Burns, and King 2013). According to a report published by the World Tourism Organization (2016), college student travelers spent \$286 billion in 2014, a figure expected to increase to around \$400 billion in 2020, representing approximately 40% growth in the overall value of the relevant market. The Generation Z population, to which most college students belong, comprises more than 65 million in the United States, which corresponds to more than 40% of all US consumers and is thus recognized as a major consumer group with enormous spending power (Priporas, Stylos, and Fotiadis 2017). These Gen Z members are essentially linked to their digital devices for various activities, such as shopping and information searches (Smith 2019). Because these younger generations respond immediately to tourism-related online content and messages, they are considered a very attractive target segment for travel product marketers (Confente and Vigolo 2018). Bravo, Catalan, and Pina (2019) found that Gen Z was a generation with very distinctive characteristics that are different from other generations in terms of reservation intentions and future intentions of information search through tourism-related websites. The fact that they are potential consumers of electrical goods and their related technological advancements and innovations make them a suitable sample for online shopping research (Yoo and Donthu 2001). One report in particular has indicated that young adults in the United States are most likely to use OTAs compared to other booking channels when purchasing tourism products, such as flights and accommodations (European Travel Commission 2020). As such, because college students are also recognized as a major consumer group in the online travel market, understanding college students' perceptions of different approaches to service recovery actions in the industry is critical to determining the actual impacts of such service recovery actions (Praesri et al. 2022).

Students participating in this survey were randomly assigned to one of four research conditions. The questionnaire was created using an online survey platform, and the survey link was emailed to the students. Once participants clicked the survey link, they were asked to read the first scenario for the assigned condition, and then they answered the first set of questions to examine their post-recovery responses. After that, they were asked to read the second scenario (with the service failure either resolved or not resolved) and then respond to a second set of questions. A total of 175 respondents participated in the experiment. After excluding two incomplete questionnaires, 173

remained, which were used to test the hypotheses. Participants who completed the survey received a small gift as a token of appreciation.

In general, the analysis of covariance (ANCOVA) test is considered robust when there are more than 25 observations in each category of the independent variable (Glass and Stanley 1970). In addition, in this case, G*Power indicated that the appropriate sample size under the conditions of this study was 128 (effect size = 0.25, power = 0.8, and $\alpha = 0.05$; Faul et al. 2007). This study used ANCOVA as the main statistical analysis technique to compare the mean values of the dependent variables (i.e., attitudes, satisfaction, and purchase intent) across groups of independent variables (i.e., active vs. passive recovery actions and high vs. low price) while controlling for covariates (i.e., OTAs usage frequency, gender, age, and race). In addition, interaction effects could also be measured by testing the joint effects of the treatments (i.e., recovery actions and price) through a factorial ANCOVA. ANCOVA is considered a powerful analysis technique that can eliminate systematic errors due to its ability to control covariates that can affect statistical results (Hair et al. 2006).

Of the students in the sample, 37 (21.4%) were male, and 135 (78.0%) were female. In 2022, the percentage of female students in universities was about 65% (NSCRC 2023), so the above percentage of females can be considered high. Regarding ethnicity, 138 (79.8%) were Caucasians. The average age of the participants was about 20 years (SD = 1.11).

3.3 | Measurement

In terms of evaluating attitudes toward OTA.com, participants were asked to respond to the following statement: “Based on this scenario, what is your overall attitude (unfavorable/favorable, bad/good, and unpleasant/pleasant) toward OTA.com?” Responses were selected from a 7-point semantic differential scale (Cronbach’s $\alpha = 0.98$; MacKenzie and Lutz 1989). Satisfaction was measured using four items on a 7-point semantic differential scale (Cronbach’s $\alpha = 0.98$; Swan and Oliver 1989). In this case, participants were asked to respond

to the following statement: “Overall, how satisfied are you with the service experience regarding OTA.com?” To evaluate purchase intentions, participants were asked to rate the following: “How likely would you be to pay for the service provided by OTA.com?” The possible responses in this case were provided on a 7-point semantic differential scale (Mohr and Webb 2005), which consisted of “very unlikely/highly likely,” “impossible/very possible,” and “no chance/certain” (Cronbach’s $\alpha = 0.97$) (Table 1). To measure the control variable, this study asked participants how often they had used OTAs to book hotels in the last 3 years. This response was evaluated through a 7-point semantic differential scale, ranging from “not very often” to “very often.” Because participants’ demographic information is also related to their attitudes and intentions (Choi and Kim 2012), their genders, ages, and races were also used as control variables.

Regarding the potential for measurement error, common method bias can occur when both independent and dependent variables are measured using the same response technique in one survey (Kock, Berbekova, and Assaf 2021). However, the occurrence of common method bias is unlikely in an experimental design study in which the independent variable is set by manipulating a specific situation and measuring the dependent variable in response to it, rather than measuring the independent variable together with the dependent variable (Dolnicar and Ring 2014; Fong et al. 2016).

4 | Results

4.1 | Manipulation Check

Manipulation checks indicated that participants exposed to active recovery actions ($M = 4.71$, $SD = 1.73$) perceived that OTA.com’s service recovery actions were significantly more active ($t = 9.19$, $p < 0.001$) than those subject to passive recovery actions ($M = 2.57$, $SD = 1.33$). Participants in the condition of high price ($M = 4.57$, $SD = 1.26$) perceived that the hotel room rate was higher ($t = 5.04$, $p < 0.001$) than those in the low price condition ($M = 3.61$, $SD = 1.26$). Regarding the believability of the stimuli, participants perceived the scenario description as believable

TABLE 1 | Measures.

Variables	Measure items	N	Mean	SD	Cronbach's α
Attitudes	Unfavorable/favorable	173	3.77	1.67	0.98
	Bad/good				
	Unpleasant/pleasant				
Satisfaction	Displeased/pleased	173	3.69	1.66	0.98
	Discontented/contented				
	Unhappy/happy				
Purchase intent	Dissatisfied/satisfied	173	3.14	1.55	0.97
	Very unlikely/very likely				
	Impossible/very possible				
	No chance/certain				

($M=5.20$, $SD=1.51$), and there were no significant differences among the conditions. In other words, the manipulations were effective in producing the intended effects. To test the normality assumption, skewness and kurtosis values were calculated. The coefficients of skewness and kurtosis fell between -1.0 and $+1.0$ and between -3.0 and $+3.0$, respectively. These results indicate that the dependent variables had a normal distribution across the levels of the independent variables (Huck 2000). Although the results of the Kolmogorov–Smirnov test for some of the dependent variables against the independent variables were statistically significant (i.e., $p < 0.05$), ANCOVA is robust to violations of the assumption of normality when group sizes are equal (Levy 1980; Olejnik and Algina 1984).

4.2 | Hypotheses Testing

The first hypothesis tested the main effects of service recovery actions on customers' attitudes toward a company. First, to examine the interaction effect of service recovery actions and price on attitudes, a two-way ANCOVA test was conducted while controlling for the impacts of the demographic variables and the frequency of OTA usage. Since there were no significant interaction effects on customer attitudes toward a company, a one-way ANCOVA test was conducted to investigate the main effects of service recovery actions on customer attitudes. Active service recovery actions generated more favorable attitudes toward a company ($M=4.66$, $SD=1.41$) than passive actions ($M=2.00$, $SD=1.04$). Because the mean difference in customer attitudes was statistically significant ($F=189.56$, $p < 0.001$), the first hypothesis was found to be supported.

H2 investigated the main effects of service recovery actions on customer satisfaction. The interaction effect of service recovery actions and price on customer satisfaction was examined using a two-way ANCOVA while controlling for the effects of the control variables. Because there were no significant interaction effects on customer satisfaction, the main effects of service recovery actions on satisfaction were examined using a one-way ANCOVA. Participants were found to be more satisfied with service recovery actions performed in an active way ($M=4.80$, $SD=1.35$) than with those executed in a passive way ($M=1.95$, $SD=1.11$). Since the mean difference in customer satisfaction was statistically significant ($F=209.01$, $p < 0.001$), H2 was considered to be supported.

Next, H3 examined the main effects of service recovery actions on purchase intentions. The results of a two-way ANCOVA showed that there were no significant interaction effects of service recovery actions and price on purchase intentions while controlling for the effects of the control variables. A one-way ANCOVA indicated that active service recovery actions engendered more favorable purchase intentions ($M=3.78$, $SD=1.53$) than passive actions ($M=1.85$, $SD=1.01$). Because the mean difference in purchase intentions was statistically significant ($F=90.01$, $p < 0.001$), H3 was supported.

For H4, the effects of service recovery actions on customers' post-recovery responses when a company recovered from a service failure were tested. A one-way ANCOVA was conducted to investigate the main effects of service recovery actions on

customers' post-recovery responses while controlling for the impacts of the control variables. Participants revealed more favorable attitudes ($M=5.48$ vs. $M=3.81$; $F=44.83$, $p < 0.001$) and purchase intentions ($M=4.60$ vs. $M=2.95$; $F=30.00$, $p < 0.001$), and they were more satisfied ($M=5.53$ vs. $M=3.76$; $F=61.28$, $p < 0.001$) with a company when it recovered from a service failure using an active approach rather than a passive strategy. Thus, H4 was confirmed.

In the case of H5, the impact of service recovery actions on customers' post-recovery responses when a company failed to address a service failure (i.e., double deviation) was tested. A one-way ANCOVA was performed to identify the main effects of the level of service recovery actions on customers' post-recovery responses while controlling for the effects of the control variables. Respondents showed more favorable attitudes ($M=3.22$ vs. $M=2.01$; $F=17.30$, $p < 0.001$) and purchase intentions ($M=2.91$ vs. $M=1.79$; $F=17.62$, $p < 0.001$), and they were more satisfied ($M=3.09$ vs. $M=1.94$; $F=15.54$, $p < 0.001$) with a company when its service recovery actions were more active rather than passive, even if a service failure was not addressed. Thus, H5 was confirmed.

Finally, H6 examined the ordinal interaction effects between service recovery actions and price on customers' post-recovery responses. The results of the two-way ANCOVA showed that there were no significant interaction effects between service recovery actions and price on customers' post-recovery responses. In addition, the results indicated that price had no significant main effects on customers' post-recovery responses. These findings show that in the event of a service failure, even if a company takes an active service recovery approach, a high or low price will not affect the customers' post-recovery responses toward the company. Thus, H6 was not supported (Table 2).

TABLE 2 | Hypothesis tests results.

Hypothesis		
H1	Active vs. passive recovery actions → attitudes	Supported
H2	Active vs. passive recovery actions → satisfaction	Supported
H3	Active vs. passive recovery actions → purchase intent	Supported
H4	(When the failure is recovered) active vs. passive recovery actions → post-recovery responses	Supported
H5	(When the failure is not fully recovered) active vs. passive recovery actions → post-recovery responses	Supported
H6	High vs. low price active vs. passive recovery actions → post-recovery responses	Not supported

5 | Discussion and Conclusions

5.1 | Discussion

Because online businesses such as OTAs are more likely to experience service failures than offline businesses due to their low levels of control and instability in service situations (McKnight and Chervany 2001; Zhao et al. 2014), OTA practitioners need to develop effective recovery actions that can alleviate negative customer reactions caused by service failure. However, not only is there a dearth of research on service failures and recovery actions related to OTAs, but there are also very few studies that have sought to identify effective recovery actions based on the various dimensions of justice theory. Moreover, studies on recovery actions related to the double deviation of OTAs remain rare, as does research on the impacts of price on the effectiveness of recovery actions. To address this gap in the literature, this study empirically investigated how recovery actions related to justice theory affect customers' post-recovery responses in initial service failure and double deviation situations and examined the role of price in this relationship.

The findings reveal that in the initial service failure situation, active recovery actions associated with procedural and interactional justice elicit significantly more favorable customers' post-recovery responses than passive recovery actions. To put it differently, these results indicate that active recovery actions involving procedural (i.e., prompt response) and interactional (i.e., sincere apology) justice for the initial service failure lead to more positive customers' responses to the OTA than passive actions (i.e., slow response and not sincere apology). Looking more specifically at the results, the mean values of the three customers' post-recovery responses to active recovery actions were all well above the median of 3.5 on each scale ($M = 4.66, 4.80$, and 3.78 for attitude toward a company, satisfaction, and purchase intentions, respectively). These numbers are significant because they were obtained without applying recovery actions related to distributive justice (e.g., discount coupons) as a recovery measure for service failures.

These results imply that, even if OTAs do not apply the recovery actions relevant to distributive justice in the event of a service failure, the application of the recovery actions related to procedural and interactional justice can achieve a significant level of positivity in customers' post-recovery responses. These findings provide empirical support to prior studies (Bacile et al. 2018; Chapman and Chapman 2016; Crisafulli and Singh 2017; Del Río-Lanza, Vázquez-Casielles, and Díaz-Martín 2009; Liu et al. 2019; Wang et al. 2011) asserting that recovery actions for procedural and interactional justice result in customers' positive recovery responses in the context of a severe service failure. When customers are subject to active recovery actions for procedural and interactional justice after the OTAs' initial service failure, they have been shown to respond positively to the recovery efforts, and this positively influences how they assess and feel emotionally about the company (i.e., attitudes and satisfaction), and whether they intend to purchase more services from OTAs.

The findings also show that active service recovery actions resulted in customers' more favorable post-recovery responses than passive recovery actions, both when the service failure was fully addressed and when it was not (i.e., double deviation). Even if

the service failure was fully addressed, customers expressed less favorable post-recovery responses toward the company when the company used passive recovery actions than when it used active ones. These results indicate that customers not only consider complete recovery from service failure to be important, but that they may also regard the efforts of recovery actions themselves leading to actual recovery to be important. In other words, to obtain customers' positive post-recovery responses to service failures, complete recovery must be achieved through active recovery actions based on procedural and interactional justice.

Even in situations in which the service failure was not fully recovered (i.e., double deviation), customers who experienced active recovery actions had more favorable post-recovery responses toward the company than those who had passive ones, albeit to a lesser extent. This also implies that customers valued a full recovery from a service failure as much as they valued the actions leading up to it. These results also show how recovery actions involving procedural and interactional justice in the service recovery process can have a significant impact on how customers react to the company that caused the failure. These findings support those of previous studies showing that since the service delivery process is not a customer experience arising at one point in time, but rather a series of interconnected service encounters, the company that caused the service failure should consider the entire service delivery process as a recovery process (i.e., service recovery journey) (Lemon and Verhoef 2016; Miller, Craighead, and Karwan 2000; Patricio, Gustafsson, and Fisk 2018; Van Vaerenbergh et al. 2019).

One of the other research objectives of this study was to investigate the moderating effects of price on recovery actions and customers' post-recovery responses. Based on utility theory and social exchange theory, it was inferred that customers who purchased rooms at relatively high prices would have higher expectations than their counterparts for recovery actions and thus would have less favorable post-recovery responses than those who purchased rooms at a lower price. However, our findings show that higher or lower prices had no effect on the relationship between recovery actions for service failures and customers' post-recovery responses.

The reason for this finding is believed to be that when a service failure occurs, customers first focus on the failure, evaluate the recovery actions taken for the failure, and determine their responses to the company based on the evaluation results. In other words, when a service failure occurs, customers do not comprehensively judge the price paid to purchase the service, the service failure, and the recovery actions for the failure together to determine whether the recovery actions for the failure are appropriate. This result can also be interpreted to mean that customers who purchase services at a low price and those who purchase a service at a high price for the same service product do not significantly differ in their expectations of recovery actions for service failures.

5.2 | Theoretical Contributions

This study provides important theoretical implications for academia with regard to the hospitality industry, especially OTAs.

First, this study demonstrates the importance of active recovery actions based on procedural and interactional justice in addressing service failures in OTAs by empirically confirming that such actions lead to more favorable post-recovery responses among customers than passive ones. No prior research has investigated the influences of recovery actions reflecting the procedural and interactional dimensions on customers' post-recovery responses in the OTAs context. Second, another major finding of this study was that active recovery actions resulted in customers' more favorable post-recovery responses compared to passive ones, both when the failure was fully recovered and when it was not (i.e., double deviation). This study shows that, in both the initial failure and the double deviation, active recovery actions based on procedural and interactional justice played an important role in generating positive responses toward the company. Not only has there been no research on how recovery actions related to procedural and interactional justice affect customers' post-recovery responses in a double deviation context, but it is also difficult to find research conducted on similar topics in OTAs environments. Finally, this study found that high and low prices had no effect on the effectiveness of recovery action. This suggests that when customers experience a service failure, they are more concerned with the failure and recovery actions, regardless of the price, rather than determining their reactions to whether the recovery actions were appropriate based on the price they paid. These findings are significant in that they suggest a new research avenue regarding the relationship between price and the effectiveness of recovery actions. Because the results differ from those reported in the existing literature, further discussion of the role of price in service failure situations is warranted.

Based on the findings, this study has several important implications for existing research. First, it introduces recovery actions that can effectively address service failure in OTAs contexts and elicit customers' positive post-recovery responses. Furthermore, the results of this study may help researchers improve their understanding of how customers perceive recovery actions relevant to procedural and interactional justice in service failure situations.

5.3 | Practical Implications

This study provides OTAs with an opportunity to gain a deeper understanding of how recovery actions for service failures affect customers' responses to a company. It also makes several practical contributions relevant to industry. First, based on the results, this study proposes that practitioners of OTAs develop a systematic recovery management system that can integrally manage all service processes related to customer recovery from the time-of-service failure to the complete recovery of that failure. This system would allow practitioners to communicate information to customers in real time through OTA apps about what they are doing to address the failure, what stage of recovery they are in, and who is responsible for the recovery process. The system would also allow customers to communicate their feedback on the recovery process to OTA management after the failure has been fully recovered. Second, the present study suggests that even if recovery from service failure continues to be delayed, the frontline employees in charge of responding to customers must

continue to provide active customer service that includes prompt communication of the progress on the incident and a sincere apology. Although the customer may be somewhat disappointed by the delay in the recovery effort, the employee's positive attitude and sincere apology can reduce the customers' negative feelings toward the company. Third, this study promotes the development of an employee training program to educate frontline service staff on the importance of recovery actions for addressing injustice along the procedural and interactional dimensions. Through such a training program, practitioners could educate service staff on the importance of issuing a prompt response and genuine apology in dealing with customers in the event of a service failure and how this will affect a customer's responses to the company. Finally, when a service failure occurs, a consistent level of active recovery must be implemented, regardless of whether the price of the service product is high or low. Even if a customer purchases a service product at a low price, the expectation of recovery actions is not different from the reverse case. Therefore, practitioners must develop general customer response procedures and train employees across these categories.

5.4 | Limitations and Further Research

Although this study has significant implications for academia and industry, it has several limitations. First, it employed scenario-based experimental designs. Although this approach is especially useful in investigating relationships between variables with high internal validity, it has the limitation that the external validity of the findings is reduced. Thus, researchers planning to investigate this topic are advised to conduct further studies in a field setting. Second, although college students have the highest frequency of internet use compared to other generations and are recognized as a major consumer group with enormous purchasing power (Jai, Burns, and King 2013; Priporas, Stylos, and Fotiadis 2017), using a student sample may have affected the external validity of the results, as students may not be fully representative of actual consumers who use OTAs. Therefore, this study recommends that future research on similar topics be conducted with a sample that is fully representative of the actual consumer group of OTAs, and that the results could therefore be generalized to other populations. Finally, the present study used the case of the double charging problem for hotel rooms booked through OTAs. This approach has the limitation of not being generalizable to other service contexts. Thus, future research may validate the results of this study in various service situations.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

Research data are not shared.

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